AN6463

Application note

Guidelines for NFC reader thermal design with ST25R210/300/500/501

Introduction

With the increase in electronic device integration and complexity, addressing the thermal performance of integrated circuits is crucial. The JEDEC Solid State Technology Association supports the semiconductor industry by proposing test methods and product standards. All JEDEC standards are available for download on the web after free registration. All STMicroelectronics datasheets state the maximum and minimum operating temperature conditions. The maximum operating temperature defines the threshold at which operating beyond this point creates a high probability of permanently damaging the device. The operating temperature range also defines the limits within which the electrical parameters are guaranteed to be within specification. In the semiconductor industry, it is common practice to specify operating temperature ranges based on either ambient temperature or junction temperature. Semiconductor vendors use either ambient temperature or junction temperature as the product operating temperature range, depending on the device and its power consumption. This document describes thermal management for ST25R NFC reader-based applications and explains how to operate the device within the specified operating temperature conditions. The list of products is given in Table 1.

Table 1. Applicable products

Type	Products
ST25 NFC readers	ST25R210, ST25R300, ST25R500, ST25R501

1 Glossary

Table 2. **Acronym definition**

Acronym	Description
DUT	Device under test; silicon package
IC	Integrated circuit
DMM	Digital multimeter
IR	Infrared
NFC	Near field communication
NMOS	N-channel metal-oxide semiconductor
PCB	Printed circuit board
PMOS	P-channel metal-oxide semiconductor
T_A	Ambient temperature, represented as T_A
T_J	Junction temperature, represented as T_J

2 Thermal systems: definitions and basic concepts

This section provides basic definitions and concepts related to thermal systems as applied to silicon-based integrated circuits (ICs).

2.1 Thermal system definitions

The following thermal definitions are specified in JEDEC standard No. 51-13:

Blackbody: A perfect radiator or absorber of infrared radiation.

Emissivity: The ratio of the radiant energy emitted by a surface to that emitted by a blackbody at the same temperature.

Heating current: Current supplied to a device under test to cause the junction temperature to rise.

Heating power: Product of heating current and heating voltage that causes the device under test junction temperature to rise.

Junction temperature: Temperature of the operating portion of a semiconductor device.

Thermal characterization parameter: Parameter that characterizes the behavior of the package. The two most commonly used thermal characterization parameters, ΨJT and ΨJB, are defined in JESD51-2 and JESD51-6. These parameters measure the temperature relationship between junction-to-top and junction-to-board. While the units are °C/W, they are not resistant, because the temperature difference is divided by the total power, not the power flowing between the two areas.

Thermal equilibrium: Condition in which no heat-producing power is applied to the device under test and the device junction temperature (T_J) is equal to the ambient temperature (T_A) in the immediate vicinity of the device. This is a thermal steady state at zero applied power.

Thermal resistance: Measure of the steady-state heat flow from a point of higher temperature to a point of lower temperature, calculated by dividing the temperature difference by the heat flow between the two points.

2.2 Thermal model

The JEDEC committee JC-15, thermal characterization techniques for semiconductor packages, provides guidelines to model thermal behavior (JESD15) and to measure, characterize, and report thermal performance (JESD51). The document JESD51-0 methodology for thermal measurement of component packages defines a standard method to determine the junction temperature under specific conditions.

Junction temperature is defined as:

$T_J = T_{J0} + \Delta T_J$

In most cases, T_{J0} is equal to the ambient temperature.

Under defined conditions for a specific environment, the change in junction temperature is calculated as:

$\Delta T_J = P_D \times R_{\theta JX} = P_D \times \theta_{JX}$

Where:

P_D is the power dissipated in the device, also referred to as heating power, stated in W.

$R_{\theta JX}$ is the thermal resistance from the device junction to the specific environment, with the alternative symbol θ_{JX}, stated in °C/W.

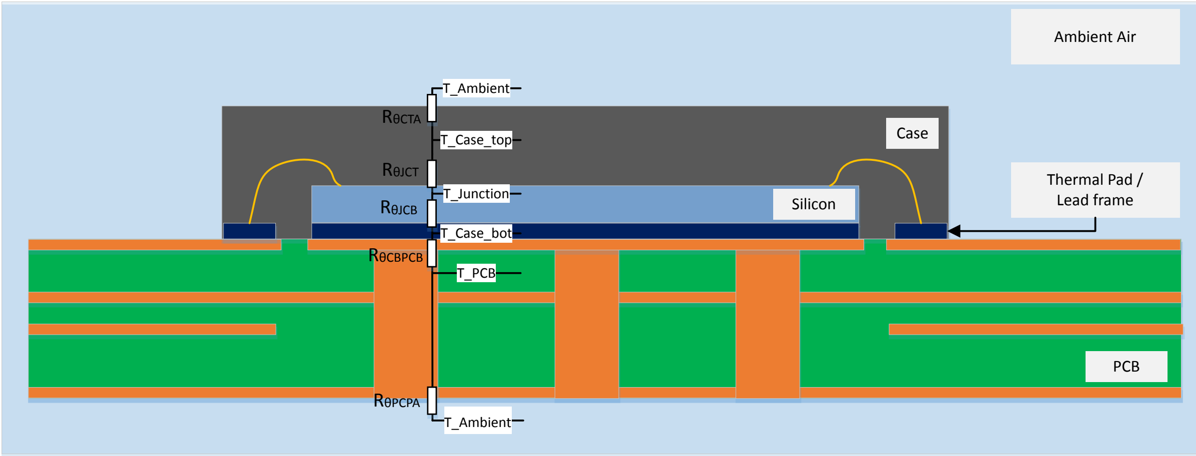
Use either ambient ($R_{\theta JA}$) or case ($R_{\theta JC}$) as the specific environment. This approach is used to set up a thermal model.

2.3 NFC reader PCB thermal model

The thermal model includes thermal resistances such as:

- $R_{\theta JCT}$ (junction to case top)
- $R_{\theta CTA}$ (case top to ambient)
- $R_{\theta JCB}$ (junction to case bottom)
- $R_{\theta CBPCB}$ (case bottom to PCB)
- $R_{\theta PCBA}$ (PCB to ambient).

Figure 1. Thermal model cross-section



The thermal resistances $R_{\theta JCT}$, $R_{\theta JCB}$, and $R_{\theta CBPCP}$ are specified for a particular silicon and package combination. $R_{\theta CBPCB}$ and $R_{\theta PCBA}$ are defined for a specific printed circuit board (PCB) and a specific environment.

For simplification, assume that there is no temperature gradient between the top and bottom sides of the PCB. This assumption is based on the fact that the top and bottom sides of the footprint are connected by several vias. These vias have high thermal conductivity.

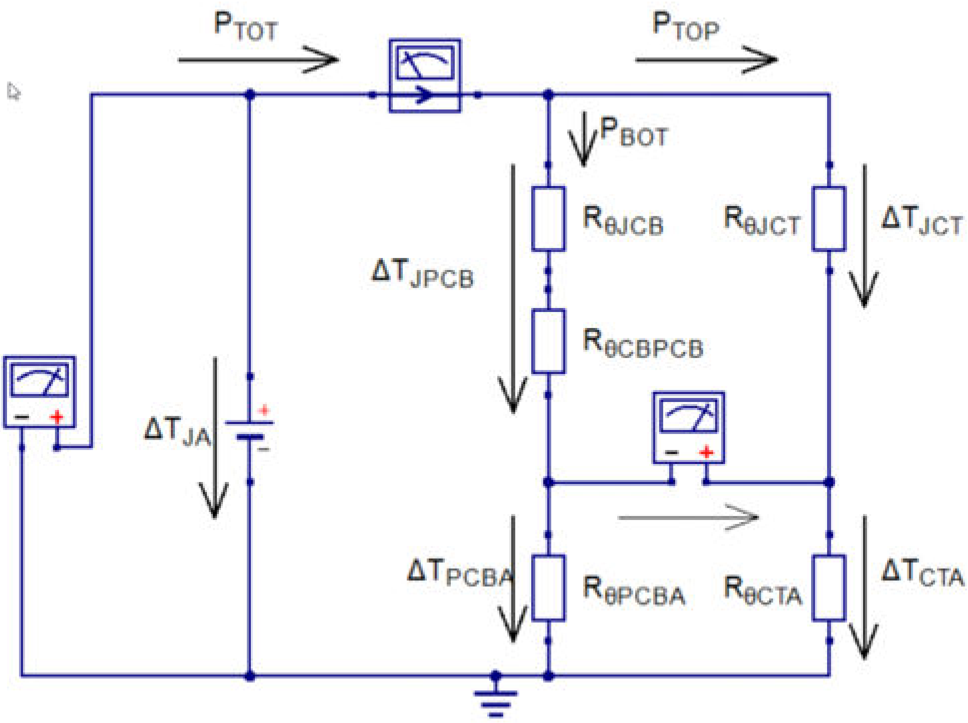
Additionally, the thermal coupling between the exposed pad of the integrated circuit (IC) and the PCB copper area provides efficient heat transfer, resulting in a very low $R_{\theta CBPCB}$.

Figure 1 shows the thermal conductivity model as an electrical model, where the silicon acts as the heat source. The heat source is represented as a voltage source, which creates a temperature difference between the junction temperature and the ambient temperature.

The heating power flows through the thermal resistance of the top and bottom case.

Figure 2 shows an equivalent electrical circuit of the thermal model in Figure 1.

Figure 2. Thermal model equivalent electrical circuit



- P_{TOT}: Total dissipated power
- P_{TOP}: Dissipated power through the top side of the package
- P_{BOT}: Dissipated power through the bottom side of the PCB
- ΔT_{JA}: Temperature difference between ambient and junction
- ΔT_{JPCB}: Temperature difference between junction, exposed pad, and PCB bottom side
- ΔT_{JCT}: Temperature difference case top side and junction
- ΔT_{JPCBA}: Temperature difference ambient and PCB bottom side
- ΔT_{JCTA}: Temperature difference ambient and case top side
- $R_{\theta JCB}$: Thermal resistance between exposed pad and junction
- $R_{\theta JCT}$: Thermal resistance between case top side and junction
- $R_{\theta CBPCB}$: Thermal resistance between exposed pad and PCB bottom side
- $R_{\theta PCBA}$: Thermal resistance between ambient and PCB bottom side
- $R_{\theta CTA}$: Thermal resistance between ambient and case top side

Table 3 shows the different thermal properties of different ST25R devices

Table 3. Thermal properties

Device	Package	$R_{\theta JCT}$ [°C/W]	$R_{\theta JCB}$ [°C/W]
ST25R501	QFN24	20.12	1.88
ST25R500	UFQFPN32	15.88	2.0
ST25R300			
ST25R210	UFQFPN28	21.42	9.2

Larger QFN packages have bigger thermal pads and more thermal vias to enhance heat dissipation from the die to the PCB. This results in a cooler thermal pad but a slightly higher temperature difference between the die junction and the pad.

Consequently, the thermal resistance junction-to-case (θ_{JC}) varies slightly between package sizes, typically by around 0.1–0.2 °C/W.

The die size impacts the thermal energy transfer between the thermal pad of the package and IC substrate. PCB design standards ensure consistent thermal performance comparisons across different package variants.

The above data has been provided by the package manufacturer and is based on JSED 51.

2.4 Power dissipation

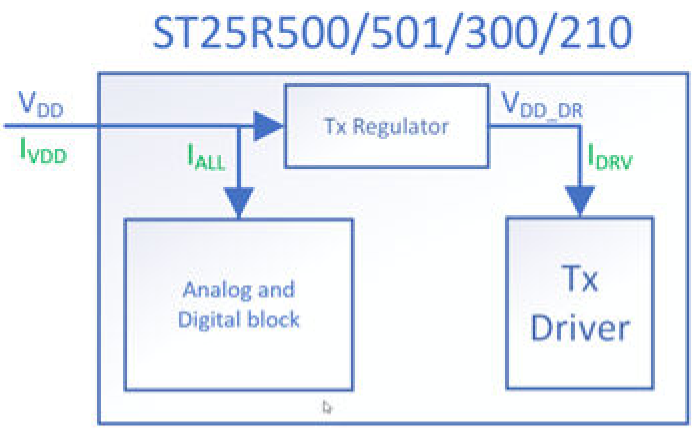
The ST25R210, ST25R300, ST25R500, and ST25R501 have three basic blocks, which produce the largest power consumption/power dissipation inside the IC.

The blocks are:

- Dissipated power of the analog and digital logic:
  - These blocks are required during operation.
  - The power consumption is calculated by:
    $P_{ALL} = V_{DD} * I_{ALL}$
  - V_{DD} is the general supply voltage of the IC. I_{ALL} is the "supply current all active," this parameter is defined in a specific datasheet of each device.
- Internal voltage regulator:
  - Determined by the following equation:
    $P_{REG} = (V_{DD} – V_{DD\_RF}) * (I_{VDD} – I_{ALL}) = V_{REG} * I_{DRV}$.
  - The voltage drop across the regulator is multiplied by the current flowing through the regulator. The current through the regulator (I_{DRV}) is calculated by measuring the overall V_{DD} current (I_{VDD}) minus the previously measured current (I_{ALL}) consumed by the analog and digital logic.
- Transmitter driver stage:
  - Determined by the following equation:
    $P_{DRV} = I^2_{DRV} * (R_{DRV\_NMOS} + R_{DRV\_PMOS})$.
  - The power dissipated in the driver is calculated by multiplying the P_{mos} and N_{mos} on-resistance of the driver stage multiplied by the square of the current flowing through the regulator.

The main power consumption blocks are illustrated in Figure 3.

Figure 3. Power consuming blocks



The driver comprises two push-pull drivers. The typical resistance is specified in the datasheet for different supply voltages.

2.5 Package temperature measurement

There are two common methods to measure the package temperature.

2.5.1 Thermocouple

Most thermocouples have a round sensor body that must be placed in contact with the flat package surface. Only a small part of the sensor body contacts the device under test (DUT) surface. This small area is heated by the DUT. The remaining surface is exposed to the ambient temperature, which cools the sensor and the DUT. Thermal compound paste is used to maximize temperature transfer between the DUT and the sensor, but it increases the DUT surface area and introduces additional thermal boundaries. A thermocouple is especially useful in closed housing environments.

2.5.2 Infrared camera module

The temperature of the PCB and IC package in this application note have been measured using a FLIR ETS320 thermal camera module.

The IR image sensor measures temperature difference very precisely, but the absolute temperature measurement has a tolerance of +/- 3 °C. There are several aspects to consider:

- Heating sources like lamps or the sun can be reflected by the DUT and cause a wrong reading.
- The IR camera module itself is a heating source. The DUT must be placed at an angle of 45 ° to the lens.
- To measure the +/- 3 °C temperature offset, the DUT must be stored at room temperature until it reaches ambient temperature. The difference between the measured temperature of the unpowered device and room temperature is the temperature offset of the camera module.
- The emissivity is the final critical parameter. It is described in the ETS320 user manual as outlined in Section Appendix A.

2.5.2.1 *IR camera temperature offset*

The previously mentioned temperature offset is easily assessed as follows:

- Store the DUT for several hours for it to reach room temperature, also known as ambient temperature.
- Measure the temperature of your DUT using the IR camera module.
- Measure the ambient temperature using a precise thermometer.

The difference between ambient temperature and measured temperature is the offset of the IR camera module.

2.6 Junction temperature measurement

There are two ways to measure the junction temperature of an IC:

2.6.1 IR camera and open IC package

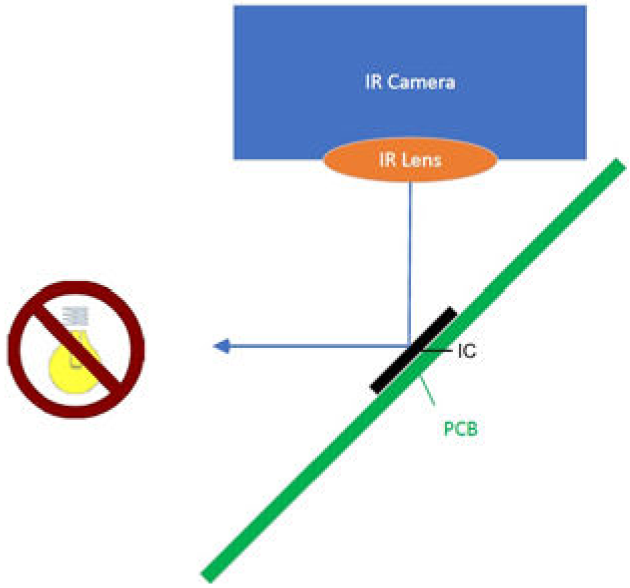
This method is like the package temperature measurement. First, the emissivity of silicon and temperature offset of the IR camera module must be found.

Then the device is turned on and the actual junction temperature increase is measured.

To avoid thermal reflections coming from the camera module or other objects near the camera module the measurement angle should be 45 ° to the measuring plane.

This is illustrated in the following figure:

Figure 4. IR camera setup



2.6.2 ESD protection diode voltage

This method uses the ESD protection diode of the RFO2 pin. It is very close to the heat source on the silicon.

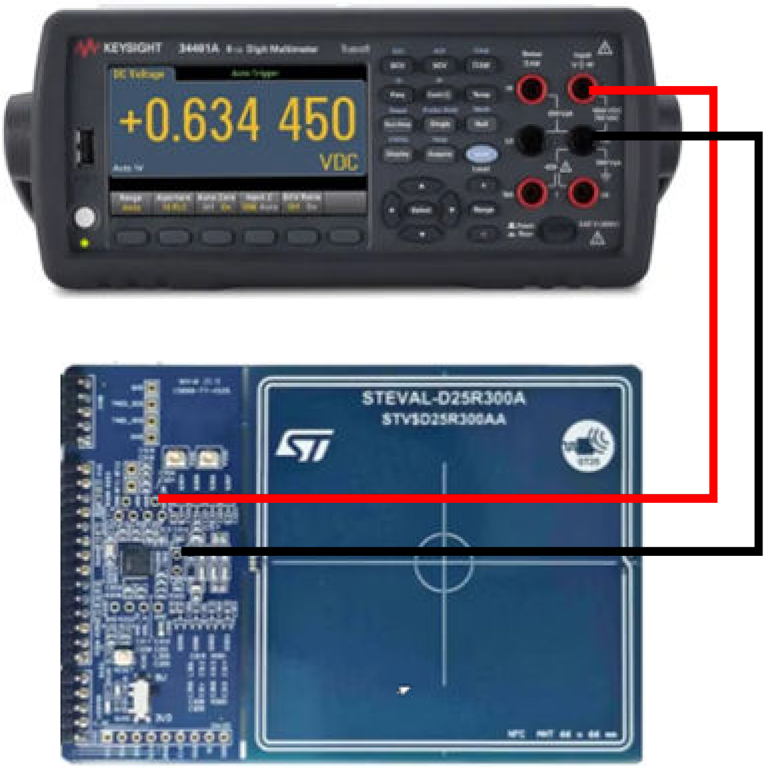
The diode measurement function of a Keysight 34461A / 34465A DMM is suitable for this purpose. The DMM forces a current of 1 mA through the diode and measures the diode voltage.

2.6.2.1 *Reference measurement*

Since the voltage varies from part to part, a characterization must be done for each device. In operation, the junction temperature increases, and the resultant diode voltage drops.

The positive connection of the DMM must be connected to GND_DR and the negative connection to RFO2. The measurement is done using the setup in Figure 5.

Figure 5. RFO1 ESD protection diode measurement setup



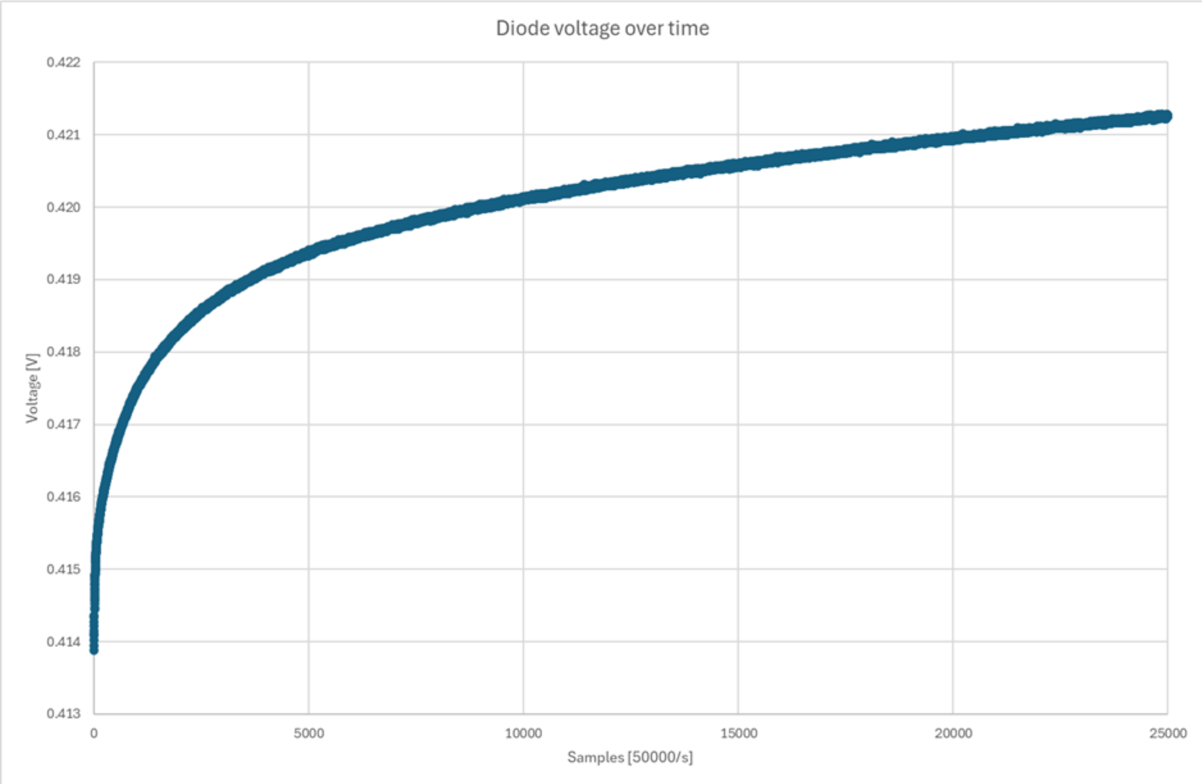
The DUT should operate in continuous wave mode to reach the highest self-heating temperature of the evaluation board.

After reaching the steady state of this self – heating, the device is switched into the Power-down mode.

At this moment, the diode voltage is recorded by the DMM.

Figure 6 shows a typical recording of such a measurement. The lowest value is chosen for the diode reference voltage.

Figure 6. Typical diode voltage measurement results



The result of this measurement is the base for the junction temperature determination in the next section.

2.6.2.2 *Junction temperature determination*

To correlate the diode voltage drop with the specific junction temperature, the device is heated by the air stream system: MPI ThermalAir TA-5000A.

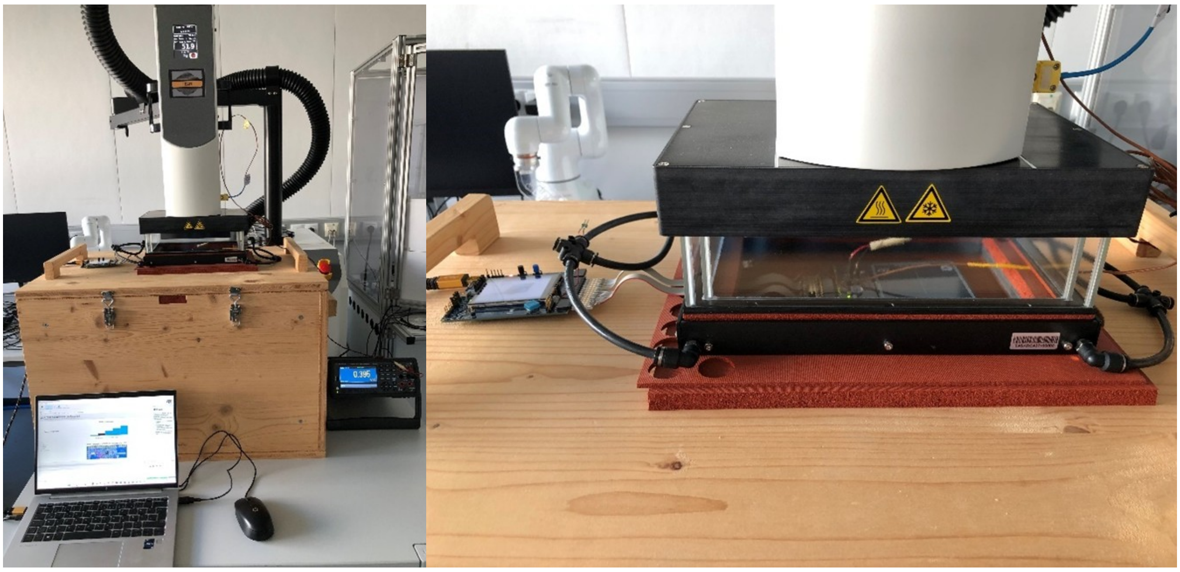
To ensure that the device is uniformly heated, the device must be kept for several minutes at the required temperature and put in power-down mode to ensure that the device does not dissipate any power during the diode voltage measurement.

Figure 7 shows the measurement setup.

The temperature of the DUT is raised until the diode measurement shows the same voltage than with the reference measurement in Section 2.6.2.1.

This means that the controlled ambient temperature of the air stream system equals the junction temperature of the IC!

Figure 7. Air stream system setup for the junction temperature determination



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3 Thermal model calculation procedure

There are six main steps required to calculate the thermal model and the junction temperature of an NFC reader device:

1. Temperature offset measurement (IR camera module)
2. Junction temperature measurements
3. Electrical parameters measurement
4. Dissipated power calculation
5. Thermal model calculation
6. Overall thermal resistance calculation

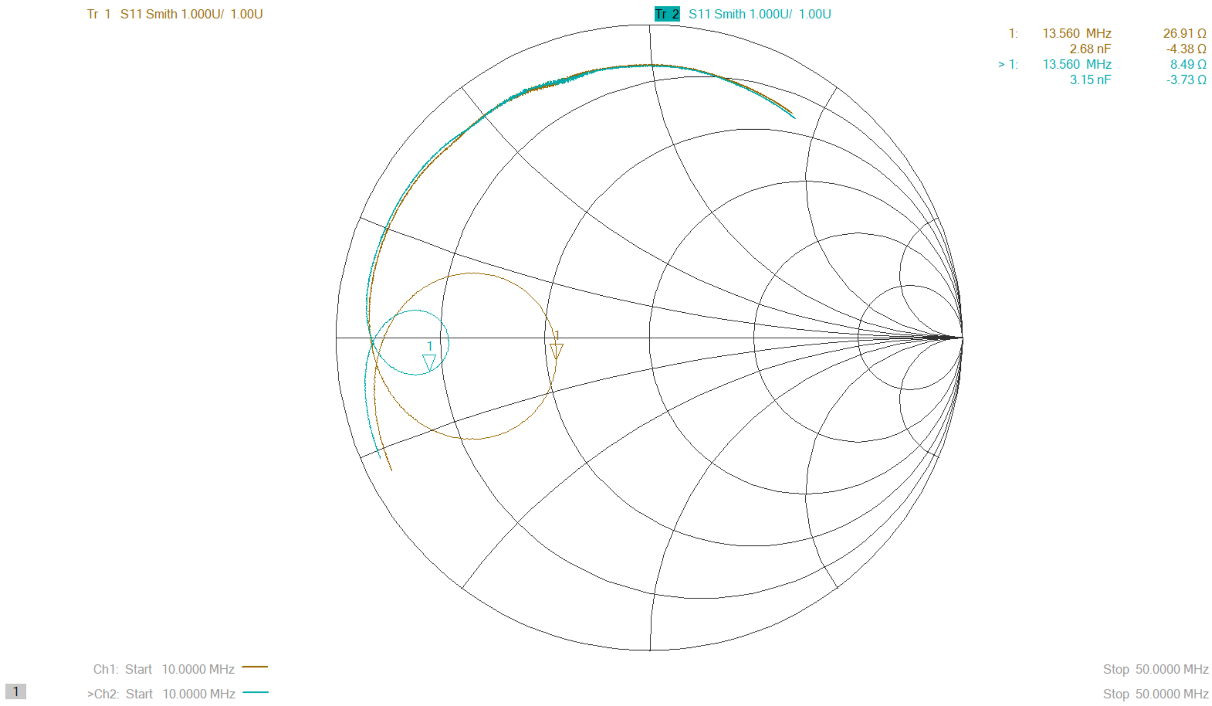
3.1 ST25R300

The STEVAL – D25R300A board is used as a reference to measure the performance of the ST25R300 with the UFQFPN32 package.

Two different matching impedances of this board are used for the thermal model investigations (see Figure 8):

- Option 1: The driver output current at this matching impedance is 133 mA (brown curve).
- Option 2: The driver output current at this matching impedance is 298 mA (blue curve).

Figure 8. **Matching impedances of the STEVAL – D25R300A board**



The STEVAL – D25R300A DUT (Package: UFQFPN32) is shown in Figure 9 and Figure 10.

The PCB consists of four layers. The dimension of the NFC part is 70 x 31.5 mm.

The QFN32 footprint has nine (3x3) thermal vias connecting the GND plane of each of the four layers together.

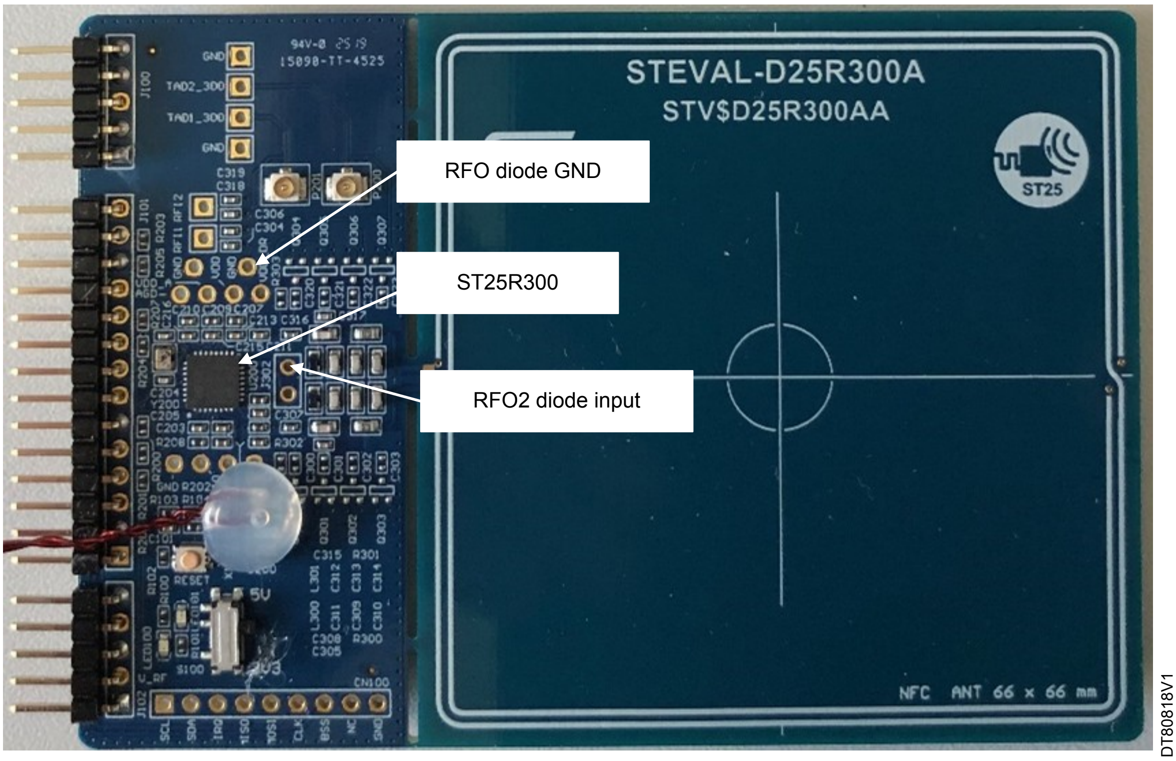


Figure 9. STEVAL – D25R300A board top view

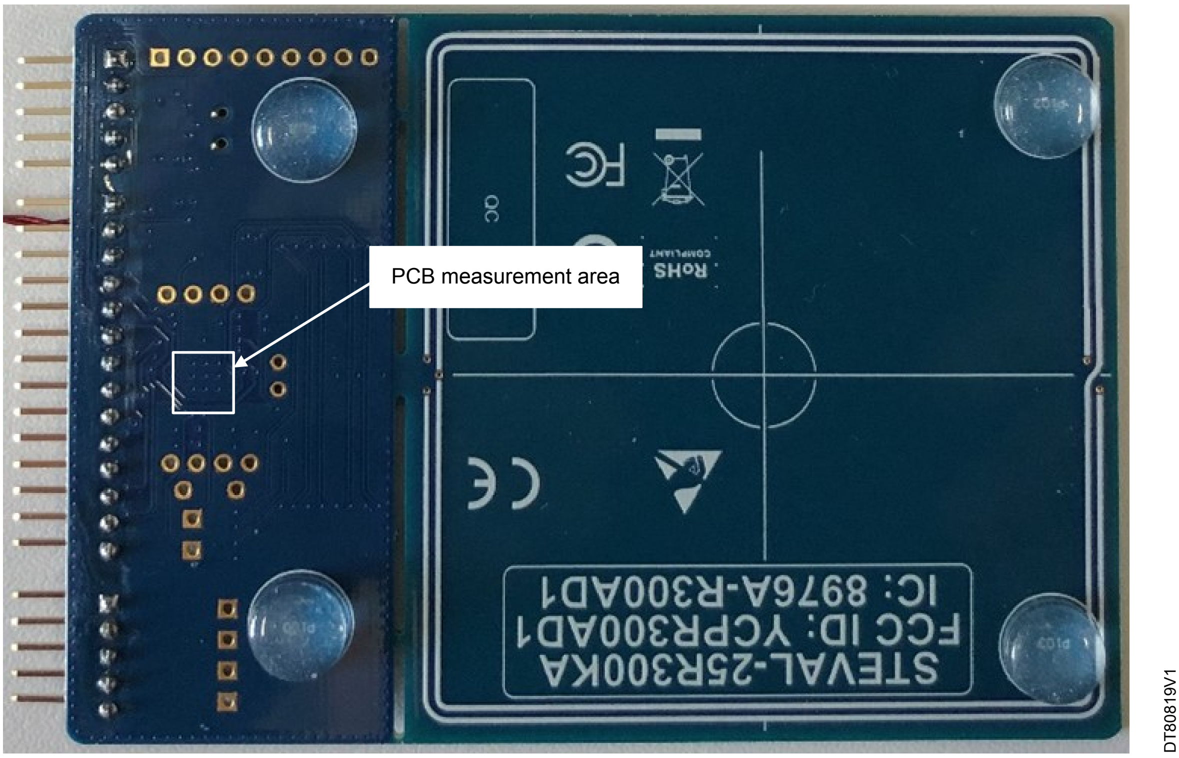


Figure 10. STEVAL – D25R300A board bottom view

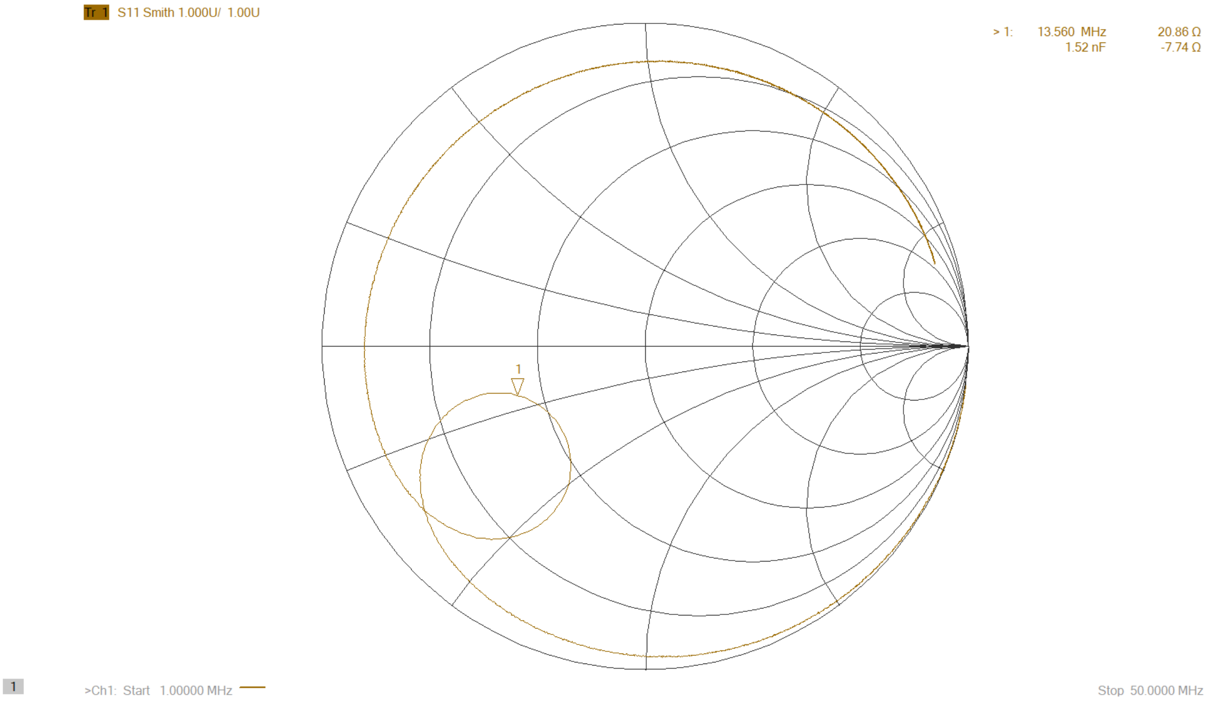
3.2 ST25R501

The ST25R501 evaluation board V2.0 is used as a reference to measure the performance of the ST25R501 with the QFN24 package.

The matching impedance of this board can be seen in Figure 11.

The driver output current at this impedance is 161 mA.

Figure 11. Matching impedance of the ST25R501 evaluation board



The ST25R501 evaluation board V2.0 DUT (Package: QFN24) is shown in Figure 12 and Figure 13.

The PCB consists of four layers. The dimension of the NFC part is 70 x 34 mm.

The QFN32 footprint has five thermal vias connecting the GND plane of each of the four layers together.

Figure 12. **ST25R501 evaluation board - Top view**

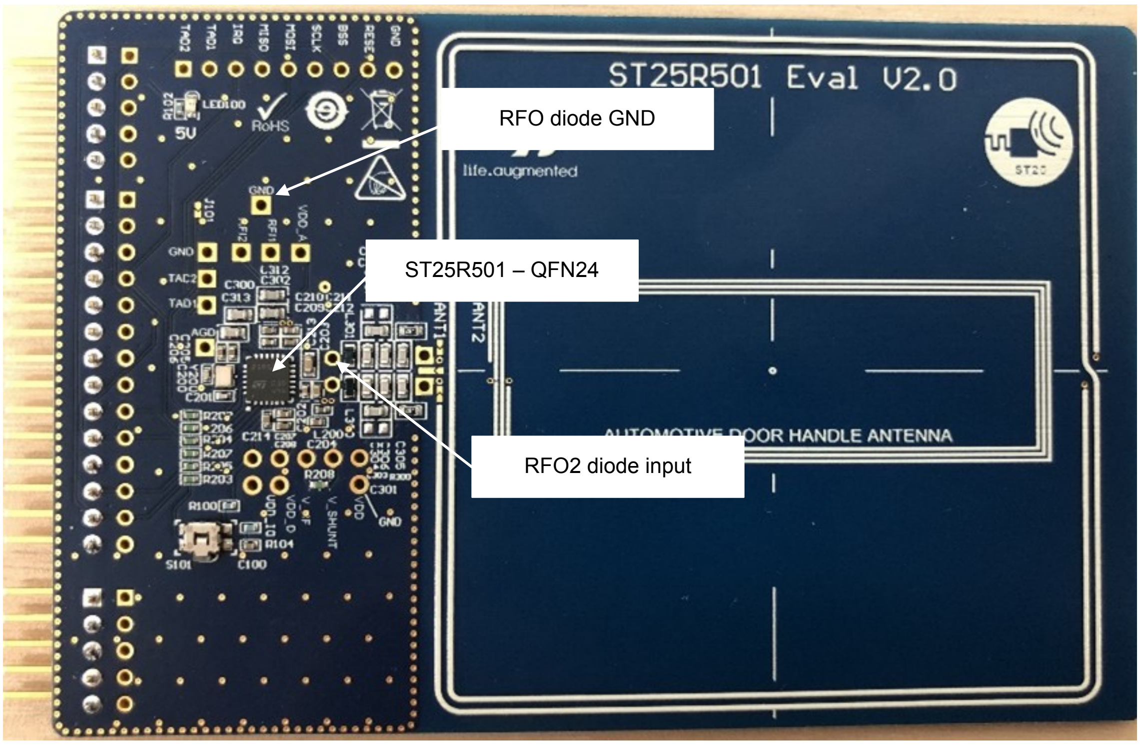
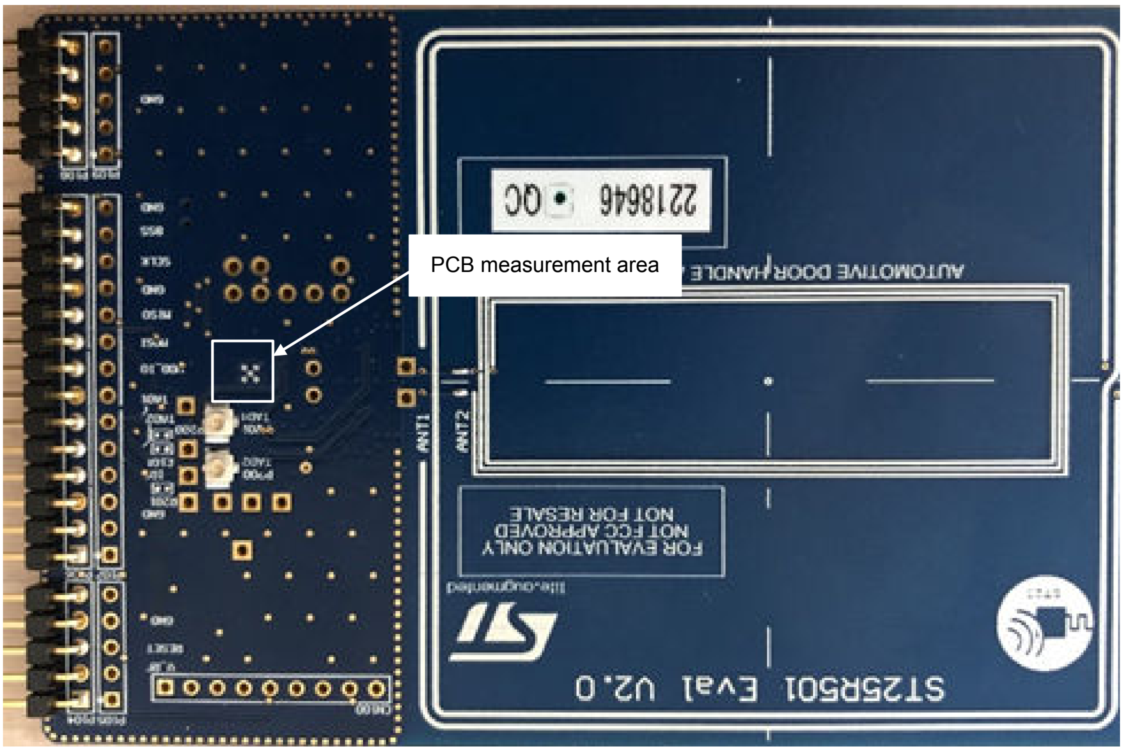


Figure 13. **ST25R501 evaluation board - Bottom view**



4 Thermal model calculation for R300/R500

This section describes the definition of the thermal model of the ST25R300/ST25R500 with the UFQFPN32 package at different driver output currents.

4.1 Thermal model definition at 133 mA

Driver output current: 133mA

PCB: STEVAL – D25R300A

The thermal model is calculated according to the procedure in Section 3.

4.1.1 Offset measurement

The first step to measure the component´s thermal dissipation is to calibrate the temperature measurement results between the IR camera module and the thermocouple.

The DUT has been stored for a sufficient amount of time at ambient temperature to ensure that the DUT temperature has reached a steady state temperature. The temperature of the DUT is measured by:

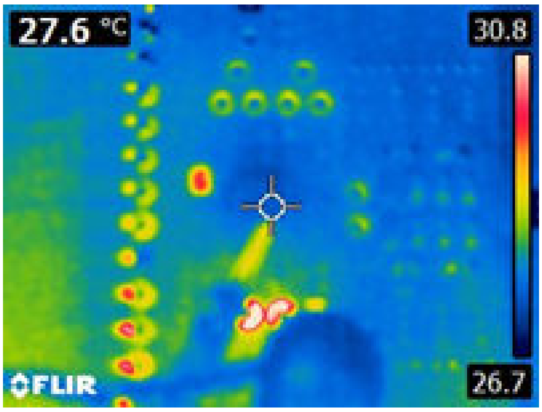
- Thermocouple as $T_{A\_thermocouple}$ = 25.6 °C
- IR camera module as T_{A_IR} = 27.6 °C as illustrated in Figure 14.

The temperature offset between the IR camera module and the thermocouple is calculated as:

$T_{Offset} = T_{A\_thermocouple} - T_{A\_IR}$ = 25.6 – 27.6 = -2 °C

Note: The thermocouple temperature is taken as the reference temperature.

Figure 14. Ambient temperature measurement (ε = 0.95)



The IR camera module has been set up to use the appropriate surface emissivity that is dependent on the measurement surface.

The temperature offset between the thermocouple and IR camera module is -2 °C. This means the IR camera module shows a temperature that is 2 °C above the actual value.

4.1.2 Junction temperature measurement

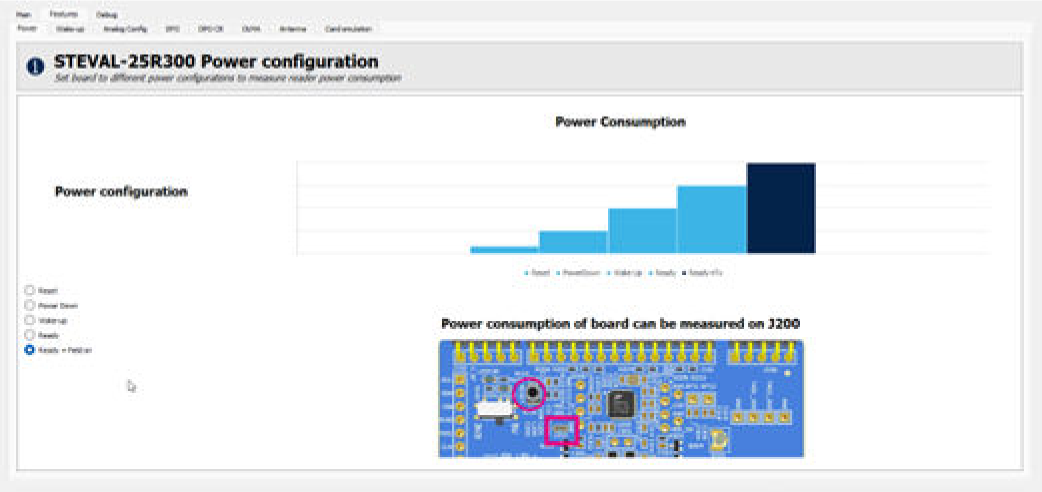
The method “ESD protection diode voltage” described in Section 2.6.2 is used in this example.

4.1.2.1 Reference measurement

The device must be put in a state where it consumes the most energy. This occurs during continuous wave output.

Figure 15 shows the GUI configuration for continuous wave output.

Figure 15. **Continuous wave output configuration**



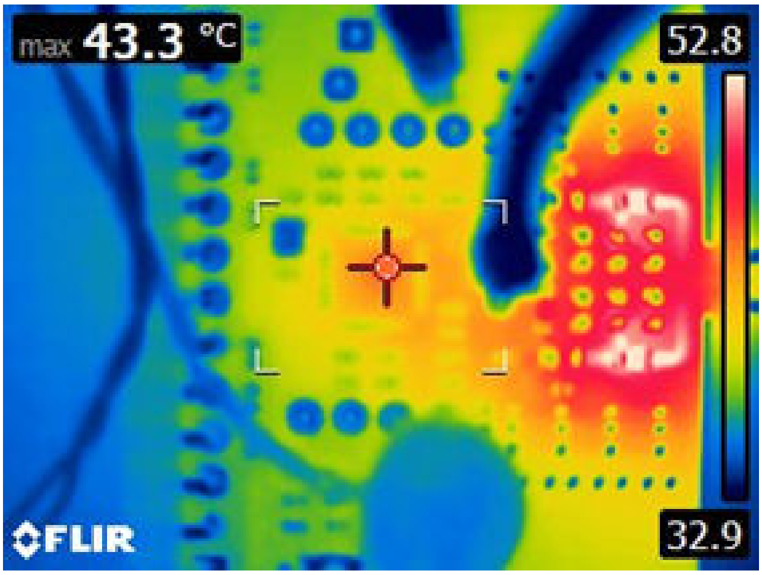
The surface temperature of the IC package is being observed until it settles to its maximum temperature.

The diode reference voltage at a top case temperature level of 41.3 °C is 407 mV.

Figure 16 shows the package temperature on the top side measured with the IR camera module. The IR camera module shows a temperature of 43.3 °C. The real temperature of the top casing is 41.3 °C (= 43.3 °C – 2 °C).

The diode reference voltage at a top case temperature level of 41.3 °C is 407 mV.

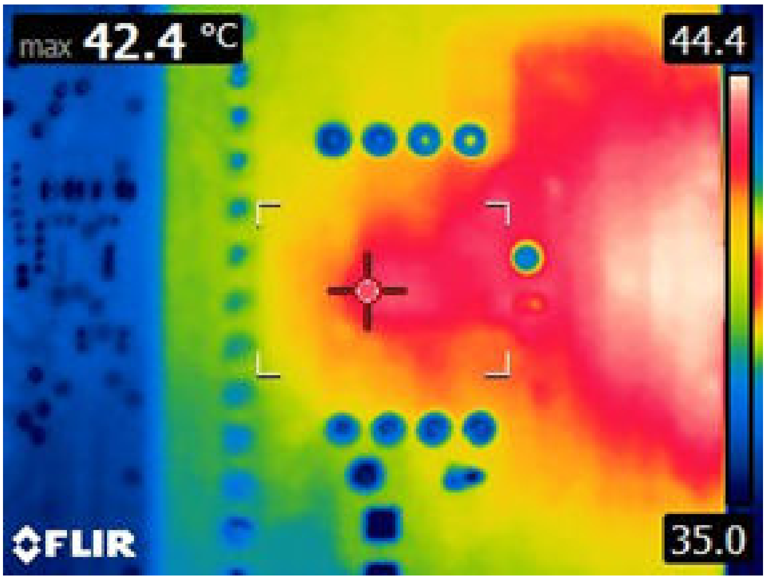
Figure 16. **Top case temperature (ε = 0.95)**



The bottom side of the PCB has been measured using the same procedure.

The emissivity (ε) has been changed to 0.9 to consideFigure 17r the surface change. The real temperature on the bottom side of the PCB is 40.4 °C (= 42.4°C – 2 °C) as illustrated in .

Figure 17. **PCB bottom temperature (ε = 0.90)**

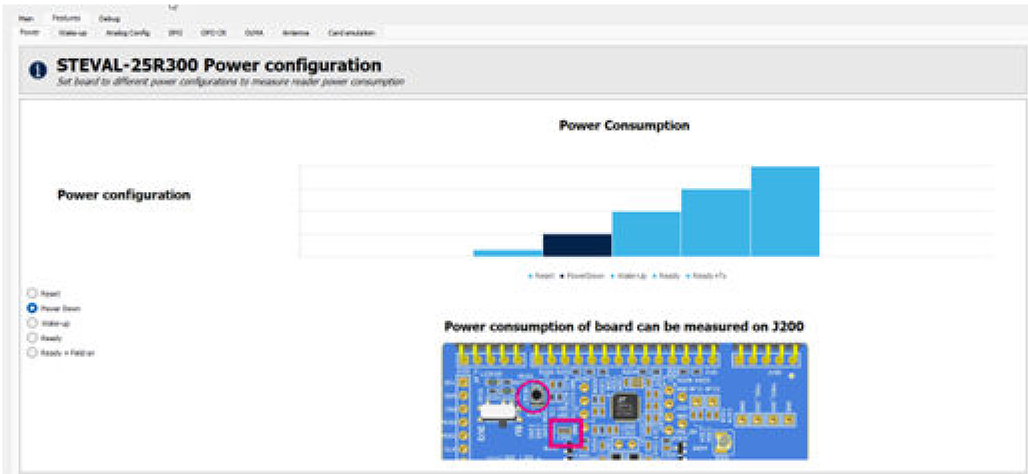


4.1.2.2 *Junction temperature determination*

The DUT must be in the power down mode for this procedure.

Figure 18 shows the GUI settings needed.

Figure 18. **Power down configuration**



The ambient temperature of the air stream system increases until the diode voltage measurement matches the value from the reference measurement (407 mV).

To achieve the same diode voltage as in the reference measurement (407 mV), increase the ambient temperature of the air stream system to 42 °C. This value also corresponds to a junction temperature of 42 °C. The difference between the junction temperature and the ambient temperature is:

$\Delta T_{JA} = T_J - T_A = 42 - 25.6 = 16.4$ °C.

4.1.3 Electrical parameter measurement

Some additional parameters are required to calculate the power dissipation that is needed for the thermal model calculation:

- V_{DD} = 4.84 V
- V_{DD_RF} = 4.44 V
- I_{VDD} = 145 mA

- I_{ALL} = 12 mA
- I_{DRV} = 133 mA
- R_{DRV_NMOS} = 1 Ω
- R_{DRV_PMOS} = 1 Ω.

4.1.4 Dissipated power calculation

The dissipated power is calculated as described in Section 2.4: Power dissipation. The previously measured electrical parameters are used in this section.

Dissipated power of the analog and digital logic:

$P_{ALL} = V_{DD} * I_{ALL} = 4.84 * 0.012 = 0.058$ W

Dissipated power of the internal voltage regulator:

$P_{REG} = (V_{DD} – V_{DD\_RF}) * I_{DRV} = (4.84 - 4.44) * 0.133 = 0.053$ W

Dissipated power of the transmitter driver stage:

$P_{DRV} = I_{DRV}^2 * (R_{DRV\_NMOS} + R_{DRV\_PMOS}) = 0.133^2*(1+1) = 0.035$ W

The total dissipated power is:

$P_{TOT} = P_{ALL} + P_{REG} + P_{DRV} = 0.146$ W

The power consumed by the device is:

$P_{IN} = V_{DD} * I_{VDD} = 4.84 * 0.145 = 0.7$ W

The output power is calculated by:

$P_{OUT} = P_{IN} - P_{TOT} = 0.554$ W

4.1.5 Thermal model calculation

The dissipated power (P_{TOT}) is emitted by the device. The power is radiated by the silicon, flowing through the previously mentioned thermal resistance between junction and package ($R_{\theta JCT}$), and package and ambient ($R_{\theta CTA}$) illustrated in Figure 1.

The calculation of the thermal model is optional.

The thermal resistance is calculated using the known junction temperature increase and total dissipated power. Nevertheless, for evaluating and optimizing the thermal performance, it is recommended to execute the calculation of the thermal model.

The two paths between bottom and top are parallel. The dissipated power splits between them and distributes according to their thermal resistance.

A lower thermal resistance results in a lower temperature increase. The previously measured temperatures are now added to the thermal model shown in Figure 19.

The temperature measured on the top package is 41.3 °C. The temperature change between case - top and ambient (ΔT_{CTA}) is calculated as:

$\Delta T_{CTA} = 41.3 – 25.6 = 15.7$ °C

The junction temperature change on the package – top (ΔT_{JCT}) is therefore given by:

$\Delta T_{JCT} = \Delta T_{JA} - \Delta T_{CTA} = 16.4 – 15.7 = 0.7$ °C

The same procedure applies for the bottom PCB temperatures:

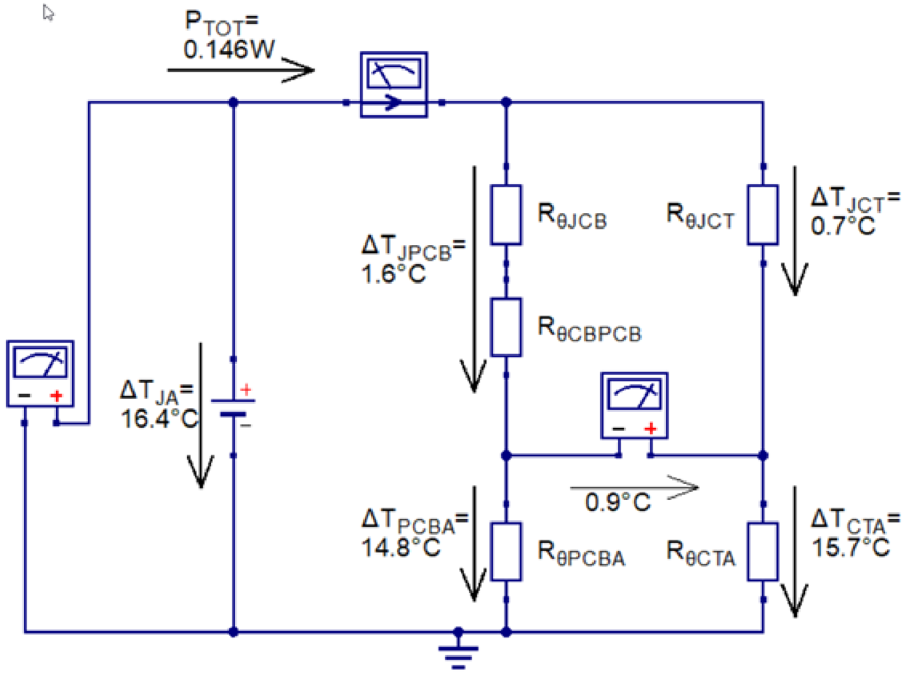
Temperature change between PCB bottom and ambient (ΔT_{PCBA}) is calculated as:

$\Delta T_{PCBA} = 40.4 – 25.6 = 14.8$ °C

The junction temperature change on the package – bottom/PCB bottom (ΔT_{JPCB}) is therefore given by:

$\Delta T_{JPCB} = \Delta T_{JA} – \Delta T_{PCBA} = 16.4 – 14.8 = 1.6$ °C

Figure 19. **Thermal model including temperature changes and PTOT**



The first step in solving the thermal model is to calculate the power flowing through $R_{\theta JCT}$ (is know from Section 2.3):

$P_{TOP} = \Delta T_{JCT} / R_{\theta JCT} = 0.7 / 15.88 = 0.044$ W

Since the power dissipated through the top case is known, $R_{\theta CTA}$ is calculated using:

$R_{\theta CTA} = \Delta T_{CTA} / P_{TOP} = 15.7 / 0.044 = 356.81$ °C/W

The power dissipated through the bottom is calculated by:

$P_{BOT} = P_{TOT} – P_{TOP} = 0.146 – 0.044 = 0.102$ W

The thermal resistance between PCB bottom side and ambient is calculated by:

$R_{\theta PCBA} = \Delta T_{PCBA} / P_{BOT} = 14.8 / 0.102 = 145.1$ °C/W

This leads to the calculation of the thermal resistance between the exposed pad and PCB bottom side ($R_{\theta CBPCB}$).

Since the amount of power dissipated through the bottom case is known, $R_{\theta CBPCB}$ is calculated using:

$R_{\theta CBPCB} = \Delta T_{\theta CBPCB} / P_{BOT} = (\Delta T_{JPCB} - \Delta T_{\theta JCB}) / P_{BOT} = (\Delta T_{JPCB} – (P_{BOT} * R_{\theta JCB})) / P_{BOT} =$

$= (1.6 – (0.102 * 2)) / 0.102 = 13.7$ °C/W

The complete thermal model can be seen in Figure 20.

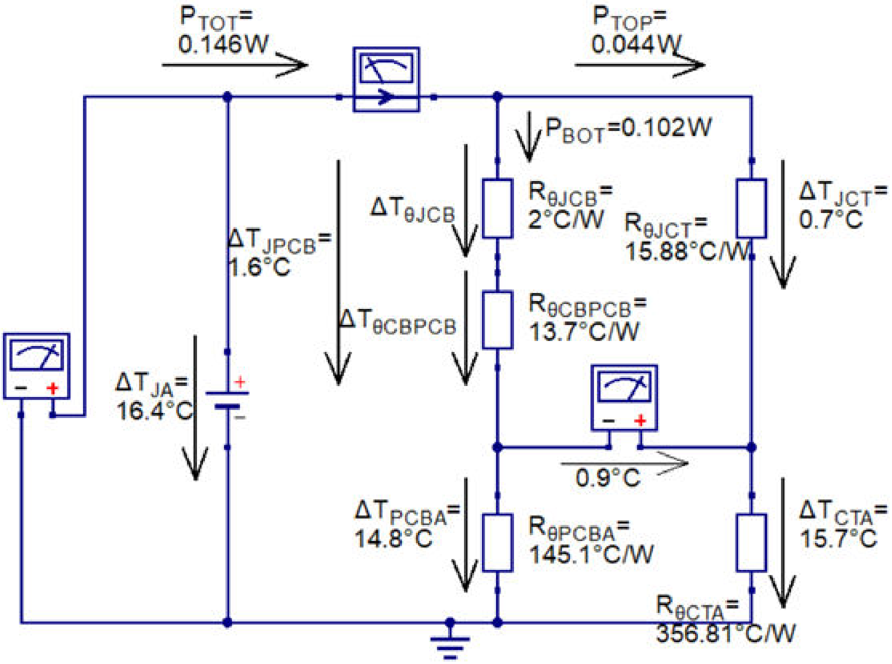


Figure 20. **Complete thermal model for the STEVALR300 with 133mA**

4.1.6 Overall thermal resistance calculation

The numbers above demonstrate the split between the power dissipated through the top-case and power dissipated through the PCB bottom side depends on the PCB and how good it dissipates the heat. The overall thermal resistance is calculated by:

$R_{\theta TH} = \Delta T_{JA} / P_{TOT} = 16.4 / 0.146 = 112.3$ °C/W

The thermal resistance $R_{\theta TH}$ is only valid for the board shown in Figure 9 and Figure 10. Using a different package, PCB layout or ambient temperature (housing or ventilation) changes $R_{\theta JCB}$, $R_{\theta CBPCB}$, and $R_{\theta JCT}$ and therefore impacts $R_{\theta TH}$.

4.2 Junction temperature estimation at 298 mA

Since the thermal model is completely solved in Section 4.1.5, the junction temperature for different operating conditions can be estimated now.

For example, looking at the thermal performance when the driver current is increased from 133 mA to 298 mA.

The following steps are required:

1. Electrical parameter measurement
2. Dissipated power calculation
3. Junction temperature estimation

4.2.1 Electrical parameters measurement

First, the electrical parameters must be defined.

- V_{DD} = 4.68 V
- V_{DD_RF} = 4.2 V
- I_{VDD} = 310 mA
- I_{ALL} = 12 mA
- I_{DRV} = 298 mA
- R_{DRV_NMOS} = 1 Ω
- R_{DRV_PMOS} = 1 Ω.

4.2.2 Dissipated power calculation

The next step is to use these electrical parameters and calculate the dissipated power for each block.

Dissipated power of the analog and digital logic:

$P_{ALL} = V_{DD} * I_{ALL} = 4.68 * 0.012 = 0.056$ W

Dissipated power of the internal voltage regulator:

$P_{REG} = (V_{DD} – V_{DD\_RF}) * I_{DRV} = (4.68 - 4.2) * 0.298 = 0.143$ W

Dissipated power of the transmitter driver stage:

$P_{DRV} = I_{DRV}^2 * (R_{DRV\_NMOS} + R_{DRV\_PMOS}) = 0.298^2 * (1+1) = 0.178$ W

The total dissipated power is:

$P_{TOT} = P_{ALL} + P_{REG} + P_{DRV} = 0.056 + 0.143 + 0.178 = 0.377$ W

The power that is consumed by the device is:

$P_{IN} = V_{DD} * I_{VDD} = 4.68 * 0.310 = 1.45$ W

The output power is calculated by:

$P_{OUT} = P_{IN} - P_{TOT} = 1.45 – 0.377 = 1.073$ W

4.2.3 Junction temperature estimation

Since the total thermal resistance of the board is determined, the difference between junction and ambient temperature is calculated by:

$\Delta T_{JA} = P_{TOT} * R_{\theta TH} = 0.377 * 112.3 = 42.3$ °C

The ΔT_{JA} states by how much the junction temperature rises, dependent on a specific ambient temperature. For example, at an ambient temperature of 25.5 °C the junction rises to:

$T_J = \Delta T_{JA} + T_A = 42.3 + 25.5 = 67.8$ °C

On the other side, a maximum ambient temperature is calculated based on the maximum junction temperature increase:

For example:

R300: $T_{A\_max} = T_{J\_max} - \Delta T_{JA} = 125 – 42.3 = 82.7$ °C

R500: $T_{A\_max} = T_{J\_max} - \Delta T_{JA} = 135 – 42.3 = 92.7$ °C

4.3 Junction temperature measurement at 298 mA

The assumptions made for V_{DD}, V_{DD_RF}, and I_{DRV} are based on real measurements of the DUT consumption sets at 298 mA.

Therefore, the estimated self-heating of the device vs. a real measurement is possible. The following steps are executed for this comparison:

1. Temperature offset measurement
2. Top and bottom temperature calculation
3. Junction temperature measurement
4. Junction temperature comparison

4.3.1 Temperature offset measurement

The same procedure as in Section 4.1.1: Offset measurement is used to determine the temperature offset between the IR camera module and the thermos couple.

$T_{A\_thermocouple} = 25.5$ °C

$T_{A\_IR} = 27.4$ °C

$T_{Offset} = T_{A\_thermocouple} - T_{A\_IR} = 25.5 – 27.4 = -1.9$ °C

The temperature offset between the thermocouple and IR camera module is -1.9 °C. This means the IR camera module shows a temperature that is 1.9 °C above the actual value.

4.3.2 Top and bottom temperature calculation

Using the thermal model defined in Section 4.1.5 the following equation can be derived:

Thermal resistance of the PCB Top side:

$R_{\theta Top} = R_{\theta JCT} + R_{\theta CTA} = 15.88 + 356.81 = 372.69$ °C/W

Thermal resistance of the PCB Bottom side:

$R_{\theta Bot} = R_{\theta JCB} + R_{\theta CBPCB} + R_{\theta PCBA} = 2 + 13.7 + 145.1 = 160.8$ °C/W

The total dissipated power is calculated using:

$P_{TOT} = \Delta T_{JA} / R_{\theta TOT}$

The dissipated power through the top case is calculated using:

$P_{TOP} = \Delta T_{JA} / R_{\theta Top}$

$P_{TOP} / P_{TOT} = (\Delta T_{JA} * R_{\theta TOT}) / (\Delta T_{JA} * R_{\theta Top}) = R_{\theta TOT} / R_{\theta Top}$

$R_{\theta TOT} = (P_{TOP} / P_{TOT}) * R_{\theta Top}$

Where:

- $R_{\theta TOT} = (R_{\theta Top} * R_{\theta Bot}) / (R_{\theta Top} + R_{\theta Bot})$
- $P_{TOP} = (P_{TOT} * R_{\theta Bot}) / (R_{\theta Top} + R_{\theta Bot}) = (0.377 * 160.8) / (372.69 + 160.8) = 0.114$ W
- $P_{BOT} = P_{TOT} - P_{TOP} = 0.377 – 0.114 = 0.263$ W

The junction case temperature is given by:

$\Delta T_{JCT} = P_{TOP} * R_{\theta JCT} = 0.114 * 15.88 = 1.8$ °C

The case top side to ambient is given by:

$\Delta T_{CTA} = P_{TOP} * R_{\theta CTA} = 0.114 * 356.8 = 40.7$ °C

The same procedure is executed for the bottom temperature.

The junction case to bottom temperature is calculated using:

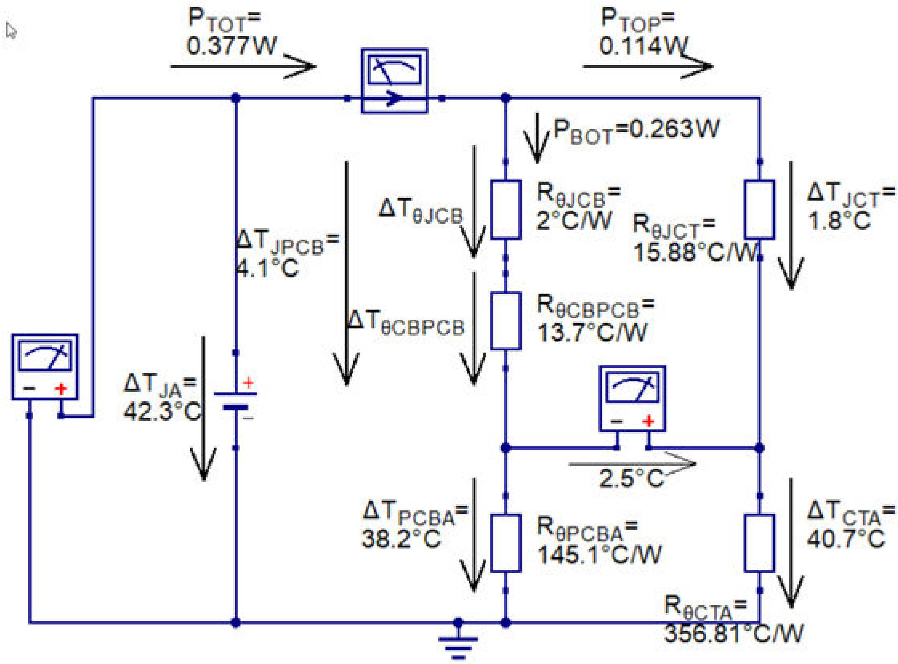
$\Delta T_{JPCB} = P_{BOT} * (R_{\theta JCB} + R_{\theta CBPCB}) = 0.263 * (2 + 13.7) = 4.1$ °C

The PCB to ambient temperature is calculated using:

$\Delta T_{PCBA} = P_{BOT} * R_{\theta PCBA} = 0.263 * 145.1 = 38.2$ °C

Figure 21 shows the complete thermal model based on these calculations.

Figure 21. Complete thermal model for the STEVALR300 with 298mA



4.3.3 Junction temperature measurement

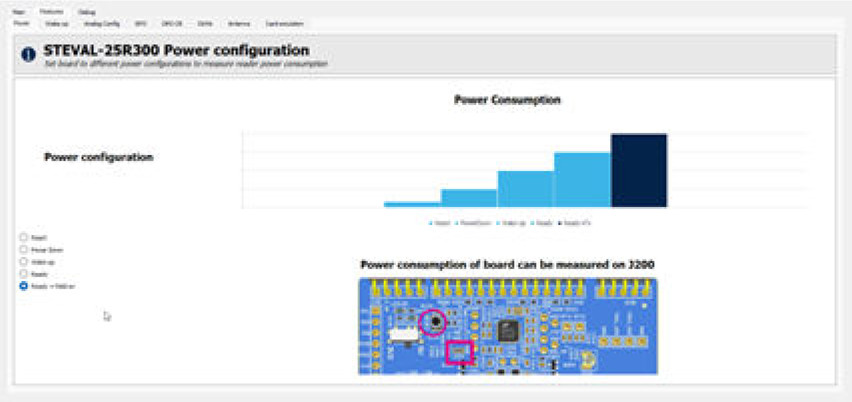
The method “ESD protection diode voltage” described in Section 2.6.2 is used in this example.

4.3.3.1 *Reference measurement*

The device must be put in a state where it consumes the most energy. This occurs during continuous wave output.

Figure 22 shows the GUI configuration for continuous wave output.

Figure 22. Continuous wave output configuration

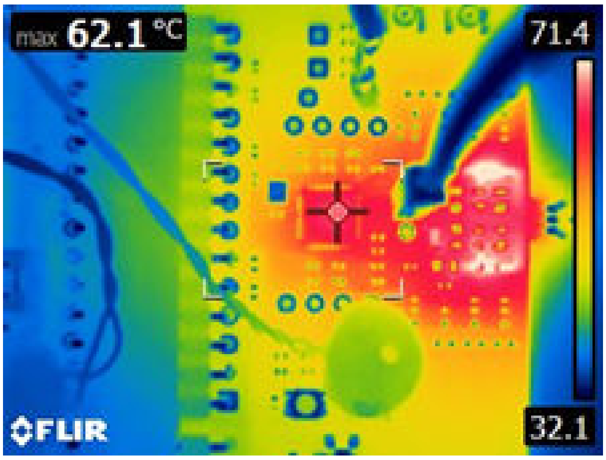


The surface temperature of the IC package is being observed until it settles to its maximum temperature.

Figure 23 shows the package temperature on the top side measured with the IR camera module. The IR camera module shows a temperature of 62.1°C. The real temperature of the top casing is 60.2°C (= 62.1°C – 1.9°C).

The diode reference voltage at a top case temperature level of 60.2°C is 384 mV.

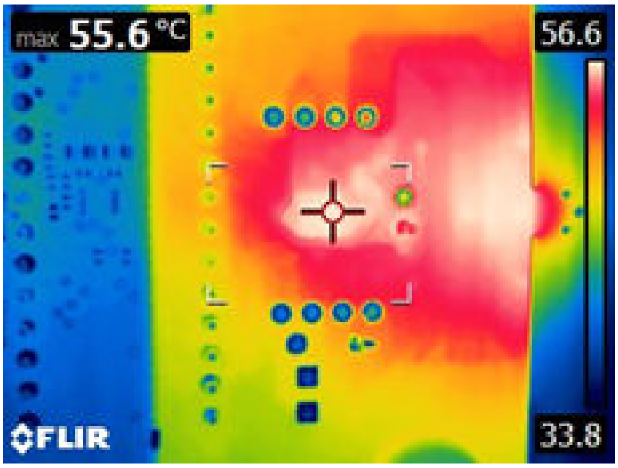
Figure 23. Top case temperature (ε = 0.95)



The bottom side of the PCB has been measured using the same procedure.

The emissivity (ε) has been changed to 0.9 to consider the surface change. The real temperature on the bottom side of the PCB is 53.7°C (= 55.6°C – 1.9°C) as illustrated in Figure 24.

Figure 24. **PCB bottom temperature (ε = 0.90)**

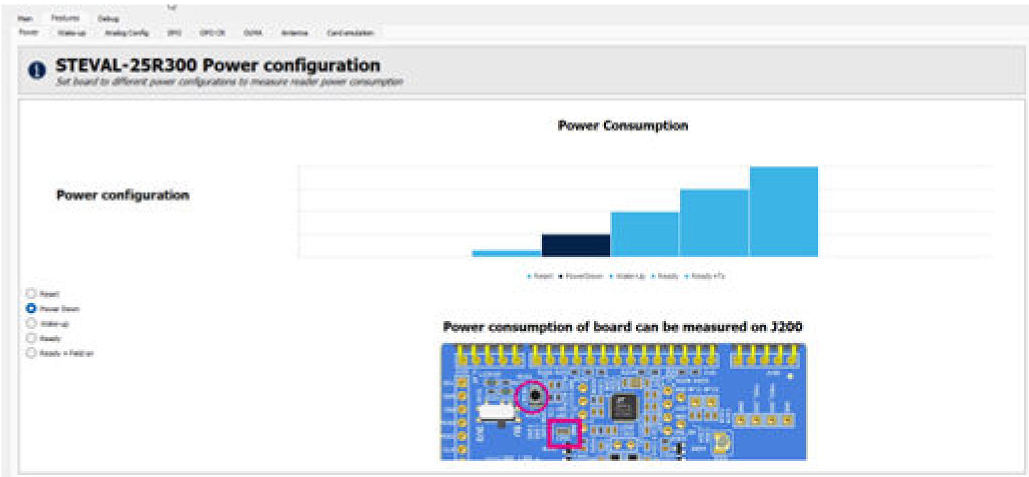


4.3.3.2 *Junction temperature determination*

The DUT must be in the power down mode for this procedure.

Figure 25 shows the GUI settings needed.

Figure 25. **Power down configuration**



The ambient temperature of the air stream system is increased as long as the diode voltage measurement shows the same value as in the reference measurement (384 mV).

To have the same diode voltage as in the reference measurement (384 mV), the ambient temperature of the air stream system must be increased to 61.2 °C.

This also equals a junction temperature of 61.2 °C.

The difference between junction and ambient temperature is:

$\Delta T_{JA} = T_J - T_A = 61.2 - 25.5 = 35.7\ °C$

4.3.4 Junction temperature comparison

Calculated value from Section 4.3.2: Top and bottom temperature calculation:

$T_{J\_calculated} = \Delta T_{JA} + T_A = 42.3 + 25.5 = 67.8\ °C$

Measured value from Section 4.3.3.2: Junction temperature determination:

$T_{J\_measured} = 61.2\ °C$

The calculated junction temperature is around 6.6°C higher than the measured one.

Note: Other heat sources, such as the matching network and EMI filter coils, can affect the package temperature measurements and, consequently, the determination of the junction temperature.

5 Thermal model calculation for R501

5.1 Thermal model definition at 161 mA

Driver output current: 161mA

PCB: ST25R501 Eval V2.0

The thermal model is calculated according to the procedure in Section 3.

5.1.1 Temperature offset measurement

The same procedure as in Section 4.1.1 is used to determine the temperature offset between the IR camera and the thermos couple.

$T_{A\_thermocouple}$ = 24.2 °C

T_{A_IR} = 26.1 °C

$T_{Offset} = T_{A\_thermocouple} - T_{A\_IR}$ = 24.2 – 26.1 = -1.9 °C

The temperature offset between the thermocouple and IR camera is -1.9 °C. This means the IR camera shows a temperature which is 1.9 °C above the actual value.

5.1.2 Junction temperature measurement

The method “ESD protection diode voltage” described in Section 2.6.2 is used in this example.

5.1.2.1 Reference measurement

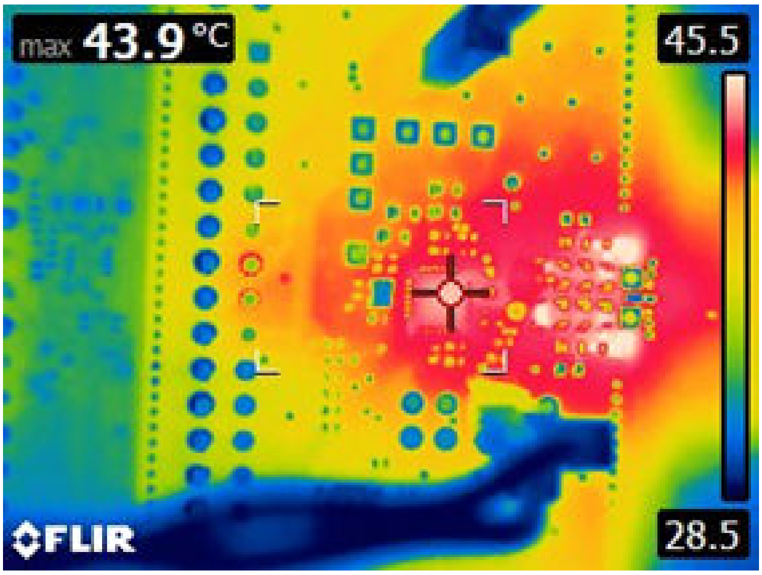
The device must be put in a state where it consumes the most energy. This occurs during continuous wave output.

The surface temperature of the IC package is being observed until it settles to its maximum temperature.

Figure 26 shows the package temperature on the top side measured with the IR camera module. The IR camera module shows a temperature of 43.9°C. The real temperature of the top casing is 42°C (= 43.9°C – 1.9°C).

The diode reference voltage at a top case temperature level of °C is 412 mV.

Figure 26. Top case temperature (ε = 0.95)



The bottom side of the PCB has been measured using the same procedure.

The emissivity (ε) has been changed to 0.9 to consider the surface change. The real temperature on the bottom side of the PCB is 40.9°C (= 42.8°C – 1.9°C) as illustrated in Figure 27.

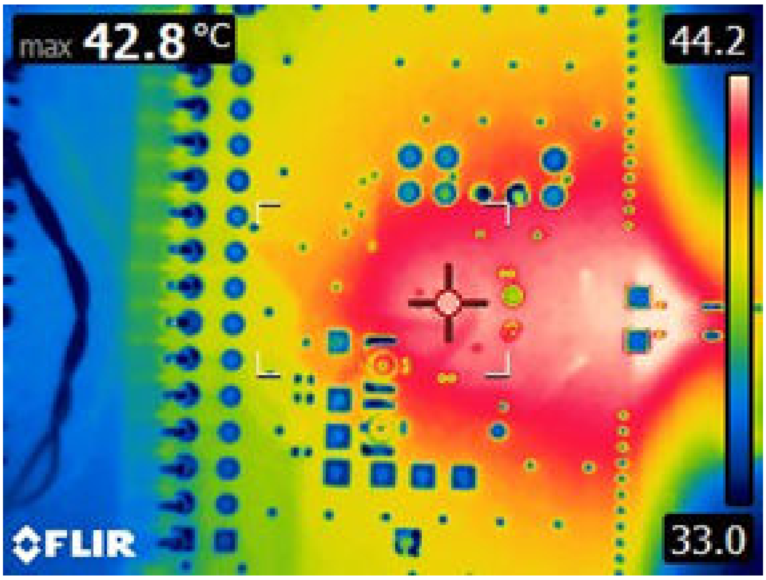


Figure 27. **PCB bottom temperature (ε = 0.90)**

5.1.2.2 *Junction temperature determination*

The DUT must be in the power down mode for this procedure.

The ambient temperature of the air stream system is increased as long as the diode voltage measurement shows the same value as in the reference measurement (412 mV).

To have the same diode voltage as in the reference measurement (412 mV), the ambient temperature of the air stream system must be increased to 43 °C.

This also equals a junction temperature of 43 °C.

The difference between junction and ambient temperature is:

$\Delta T_{JA} = T_{Junction} – T_A = 43 – 24.2 = 18.8$ °C

5.1.3 Electrical parameter measurement

Some additional parameters are required to calculate the power dissipation that is needed for the thermal model calculation:

- $V_{DD} = 4.8$ V
- $V_{DD\_RF} = 4.4$ V
- $I_{VDD} = 173$ mA
- $I_{ALL} = 12$ mA
- $I_{DRV} = I_{VDD} - I_{ALL} = 173 - 12 = 161$ mA
- $R_{DRV\_NMOS} = 0.63\ \Omega$
- $R_{DRV\_PMOS} = 0.63\ \Omega$.

5.1.4 Dissipated power calculation

The dissipated power is calculated as described in Section 2.4: Power dissipation. The previously measured electrical parameters are used in this section.

Dissipated power of the analog and digital logic:

$P_{ALL} = V_{DD} * I_{ALL} = 4.8 * 0.012 = 0.058$ W

Dissipated power of the internal voltage regulator:

$P_{REG} = (V_{DD} – V_{DD\_RF}) * I_{DRV} = (4.8 - 4.4) * 0.161 = 0.064$ W

Dissipated power of the transmitter driver stage:

$P_{DRV} = I_{DRV}^2 * (R_{DRV\_NMOS} + R_{DRV\_PMOS}) = 0.161^2*(0.63+0.63) = 0.033$ W

The total dissipated power is:

$P_{TOT} = P_{ALL} + P_{REG} + P_{DRV} = 0.155$ W

The power consumed by the device is:

$P_{IN} = V_{DD} * I_{VDD} = 4.8 * 0.173 = 0.83$ W

The output power is calculated by:

$P_{OUT} = P_{IN} - P_{TOT} = 0.68$ W

5.1.5 Thermal model calculation

The dissipated power (P_{TOT}) is emitted by the device. The power is radiated by the silicon, flowing through the previously mentioned thermal resistance between junction and package ($R_{\theta JCT}$), and package and ambient ($R_{\theta CTA}$) illustrated in Figure 1.

The calculation of the thermal model is optional.

The thermal resistance is calculated using the known junction temperature increase and total dissipated power. Nevertheless, for evaluating and optimizing the thermal performance, it is recommended to execute the calculation of the thermal model.

The two paths between bottom and top are parallel. The dissipated power splits between them and distributes according to their thermal resistance.

A lower thermal resistance results in a lower temperature increase. The previously measured temperatures are now added to the thermal model shown in Figure 28.

The temperature measured on the top package is 42 °C. The temperature change between case - top and ambient (ΔT_{CTA}) is calculated as:

$\Delta T_{CTA} = 42 – 24.2 = 17.8$ °C

The junction temperature change on the package – top (ΔT_{JCT}) is therefore given by:

$\Delta T_{JCT} = \Delta T_{JA} - \Delta T_{CTA} = 18.8 – 17.8 = 1$ °C

The same procedure applies for the bottom PCB temperatures:

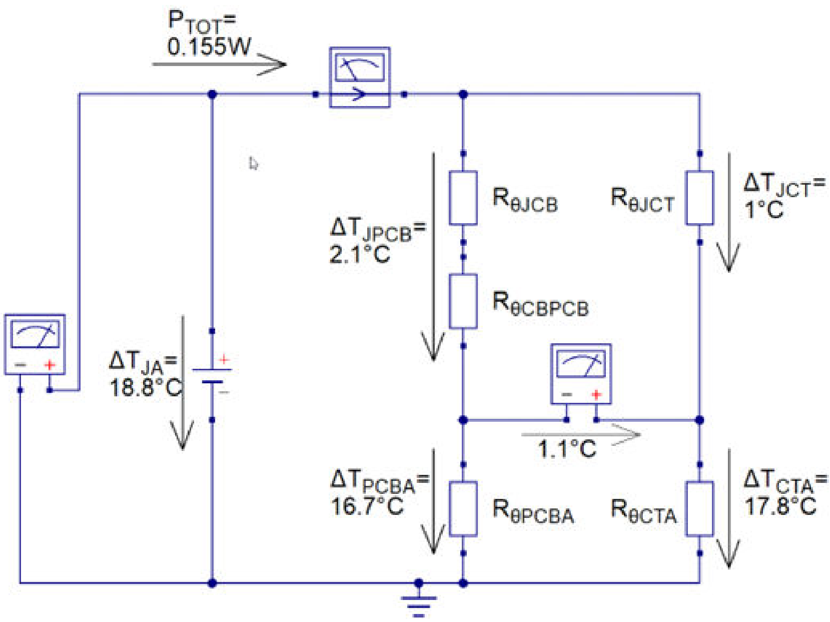
Temperature change between PCB bottom and ambient (ΔT_{PCBA}) is calculated as:

$\Delta T_{PCBA} = 40.9 – 24.2 = 16.7$°C

The junction temperature change on the package – bottom/PCB bottom (ΔT_{JCBPCB}) is therefore given by:

$\Delta T_{JPCB} = \Delta T_{JA} – \Delta T_{PCBA} = 18.8 – 16.7 = 2.1$ °C

Figure 28. Thermal model including temperature changes and PTOT



The first step in solving the thermal model is to calculate the power flowing through $R_{\theta JCT}$ (is know from Section 2.3):

$P_{TOP} = \Delta T_{JCT} / R_{\theta JCT} = 1 / 20.12 = 0.05$ W

Since the power dissipated through the top case is known, $R_{\theta CTA}$ is calculated using:

$R_{\theta CTA} = \Delta T_{CTA} / P_{TOP} = 17.8 / 0.05 = 356$ °C/W

The power dissipated through the bottom is calculated by:

$P_{BOT} = P_{TOT} – P_{TOP} = 0.155 – 0.05 = 0.105$W

The thermal resistance between PCB bottom side and ambient is calculated by:

$R_{\theta PCBA} = \Delta T_{PCBA} / P_{BOT} = 16.7 / 0.105 = 159$°C/W

This leads to the calculation of the thermal resistance between the exposed pad and PCB bottom side ($R_{\theta CBPCB}$).

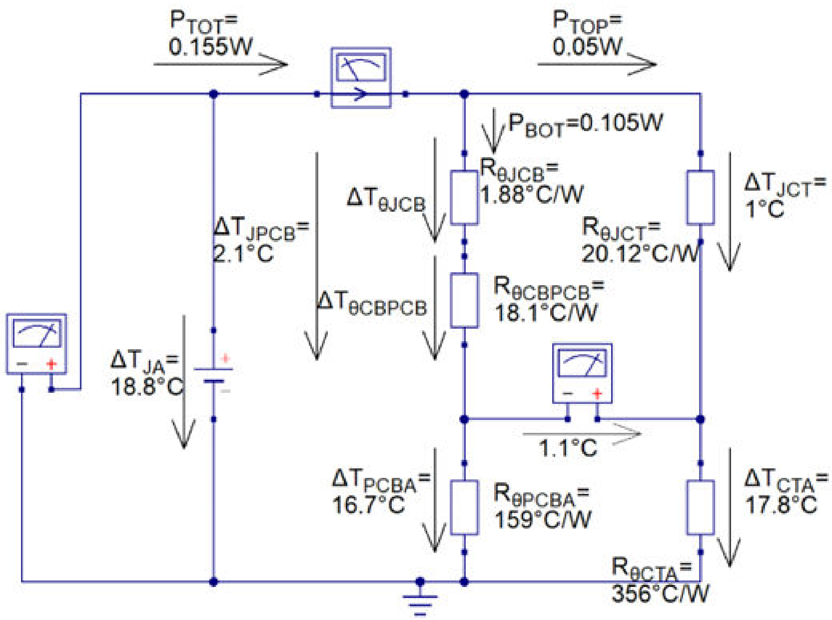
Since the amount of power dissipated through the bottom case is known, $R_{\theta CBPCB}$ is calculated using:

$R_{\theta CBPCB} = \Delta T_{\theta CBPCB} / P_{BOT} = (\Delta T_{JPCB} - \Delta T_{\theta JCB}) / P_{BOT} = (\Delta T_{JPCB} – (P_{BOT} * R_{\theta JCB})) / P_{BOT} =$

$= (2.1 – (0.105 * 1.88)) / 0.105 = 18.1$ °C/W

The complete thermal model can be seen in Figure 29.

Figure 29. Complete thermal model



5.1.6 Overall thermal resistance calculation

The numbers above demonstrate the split between the power dissipated through the top-case and power dissipated through the PCB bottom side depends on the PCB and how good it dissipates the heat. The overall thermal resistance is calculated by:

$R_{\theta TH} = \Delta T_{JA} / P_{TOT} = 18.8 / 0.155 = 121.3$°C/W

The thermal resistance $R_{\theta TH}$ is only valid for the board shown in Figure 12 and Figure 13. Using a different package, PCB layout or ambient temperature (housing or ventilation) changes $R_{\theta JCB}$, $R_{\theta CBPCB}$, and $R_{\theta JCT}$ and therefore impacts $R_{\theta TH}$.

6 Conclusion

This document describes the following:

- Thermal system definitions and basic concepts
- The thermal model calculation procedure
- Examples of thermal model calculation for NFC reader products: ST25R300, ST25R500, and ST25R501

The information described above is necessary to determine the junction temperature of the reader product based on the following factors:

- PCB design
- IC package
- Die size
- Ambient temperature

Note: Other heat sources, such as the matching network and EMI filter coils, can affect the package temperature measurements and, consequently, the determination of the junction temperature.

Appendix A Emissivity

Emissivity is one of the most important parameters in thermal imaging. It describes the ability of a material to emit radiation compared to a perfect blackbody of the same temperature.

Normally the emissivity ranges from 0.1 to 0.95, where a highly polished surface falls below 0.1, while an oxidized or painted surface has a higher emissivity. Human skin exhibits an emissivity 0.97 to 0.98.

Further information about emissivity is available in the user manual of the ETS320 IR camera: (ETS320 User manual - #T810252; r. AD/43675/43675; en-US – page 46).

This user manual also describes a method to determine the emissivity factor of materials outlined as follows:

1. Select a place to put the sample.
2. Determine and set the reflected apparent temperatures according to the previous step.
3. Put a piece of electrical tape with known high emissivity on the sample.
4. Heat the sample at least 20 K above room temperature. Ensuring the heating is evenly distributed.
5. Focus and autoadjust the camera module and freeze the image.
6. Adjust level and span for the best image brightness and contrast.
7. Set the emissivity to that of the tape (usually 0.97).
8. Measure the temperature of the tape using one of the following measurement functions:

- Isotherm (helps to determine both the temperature and how evenly the sample is heated)
- Spot (simpler)
- Box average (good for surfaces with varying emissivity)

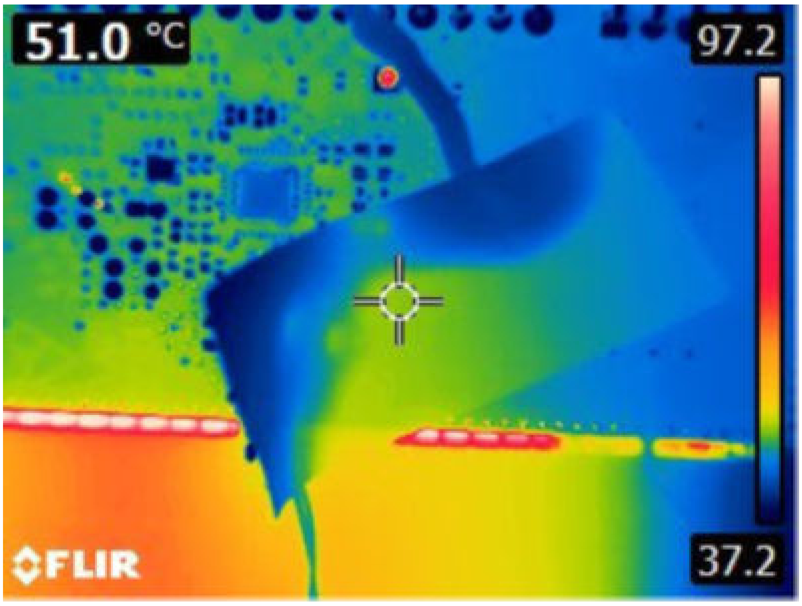
A sample measurement is illustrated in Figure 30.

9. Make a note of the temperature.
10. Move the measurement function to the sample surface.
11. Change the emissivity setting until the same temperature reading is achieved as the previous measurement.
12. Make a note of this emissivity.

To carry out this procedure, an unpowered ST25R3916-DISCO board is heated to 51°C.

The ST25R3916-DISCO board is assembled with an ST25R3916 QFN package. The top side of the package is removed to measure the actual silicon temperature.

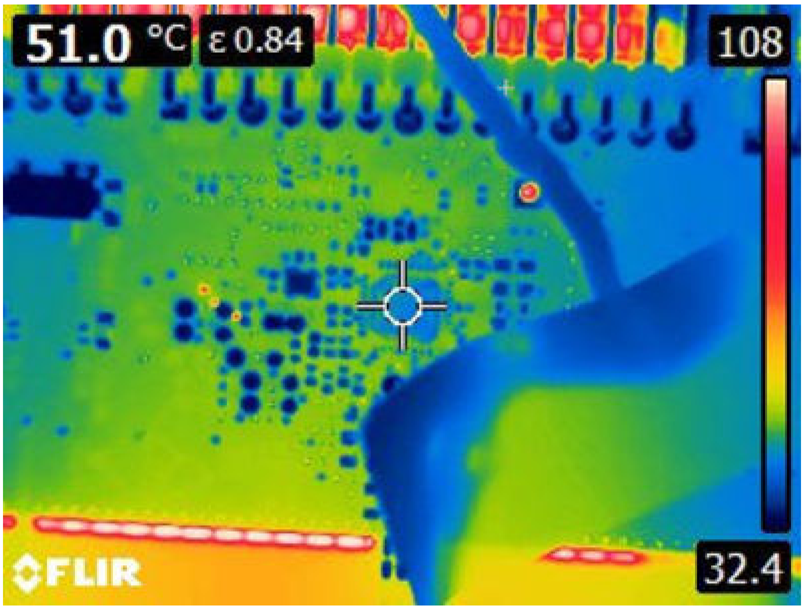
Figure 30. Measurement on plastic tape ϵ = 0.97



Note: The cross represents the measuring point that is on the electric tape

Then the focus spot is changed to the top side of the opened IC and the emissivity setting of the camera module is altered until the same temperature is displayed as illustrated in Figure 31.

Figure 31. **Measurement on the silicon top, adjust ε = 0.84**



This procedure must be repeated until the ε for each material is determined. In Table 4, the following materials have been characterized:

Table 4. **Temperature emissivity coefficients**

Material	ε
PCB surface (solder resist)	0.9
IC VFQFPN32 package	0.95
IC WLCSP package	0.97

Revision history

Table 5. Revision history

Date	Revision	Changes
01-Mar-2026	1	Initial release

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